BLACKJACK GAME APPLICATION

Course Name: CPR 210 Mobile Application Development

Student Name: Kenan Esat Şen

Student Number: 202246031

Instructor: Lect. Efe Çiftci

University: Çankaya University

Department: Computer Programming

Date: 31.12.2025

Project Overview

- **Blackjack Game Application** is a mobile card game developed for the Android platform.
  The application is based on the classic Blackjack rules and provides an interactive gaming experience for the user.
- It is developed using **Android Studio** and **Jetpack Compose**.
- The player plays Blackjack against the **Dealer**.
- Real Blackjack rules are implemented (**Hit, Stand, Bust, Blackjack, Push**).
- Game results are displayed **instantly on the screen**.
- The application features a **modern and simple user interface**.

Blackjack Rules

Objective

- The objective of the game is to reach a hand value **as close to 21 as possible** without exceeding it.

Card Values

- Number cards (**2–10**) are worth their face value
- **Jack (J), Queen (Q), King (K)** are worth **10**
- **Ace (A)** can be worth **1 or 11**, depending on the hand value

Gameplay Rules

- The player starts with **two cards**
- The dealer also starts with **two cards**, with one card hidden
- The player can choose:
  - **Hit** → draw another card
  - **Stand** → end the turn
- If the player's hand exceeds **21**, the player **loses (Bust)**

Dealer Rules

- The dealer must draw cards until the hand value is **17 or higher**
- After the dealer finishes, hand values are compared

Winning Conditions

- **Blackjack** → 21 with the first two cards
- **Player Win** → player’s hand is higher than the dealer’s
- **Dealer Win** → dealer’s hand is higher than the player’s
- **Push** → both hands have the same value
- **Bust** → hand value exceeds 21

Technologies Used

- **Development Tools**
- **Android Studio**
  Used as the main development environment for building and testing the application.
- **Programming Language**
- **Kotlin**
  The application is developed using Kotlin, which is the official programming language for Android.
- **UI Framework**
- **Jetpack Compose**
  A modern UI toolkit used to build the user interface with a declarative approach.

◈ Architecture & State Management

Composable functions

State management with (remember) and (mutableStateOf)

Single Activity architecture

◈ Design

Material 3 components

Dark theme with a clean and modern layout

How the Game Works

◈ Game Start

The game starts when the player presses the **Deal** button.

Both the player and the dealer receive **two cards**.

One of the dealer’s cards is hidden.

◈ Player Turn

The player chooses between:

Hit → draw another card

Stand → end the turn

If the player's hand exceeds **21**, the game ends with a **Bust**.

◈ Dealer Turn

After the player stands or reaches 21, the dealer starts playing.

The dealer draws cards until the hand value is **17 or higher**.

◈ Game Result

The hand values of the player and the dealer are compared.

The result is displayed on the screen:

Player Win

Dealer Win

Push

◈ New Round

The player can start a new game by pressing **New Round**.

◈ Game Start

Initial game screen before starting the round

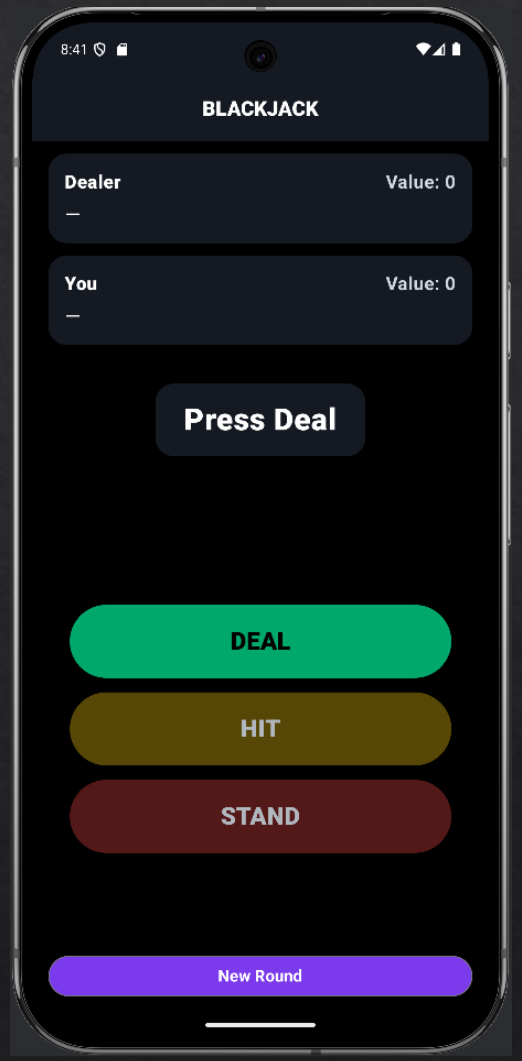
Player presses **Deal** to start the game

◈ New Round

Allows the player to **reset the game state**

Clears previous hands and starts a **new round**

Enables continuous gameplay without restarting the application

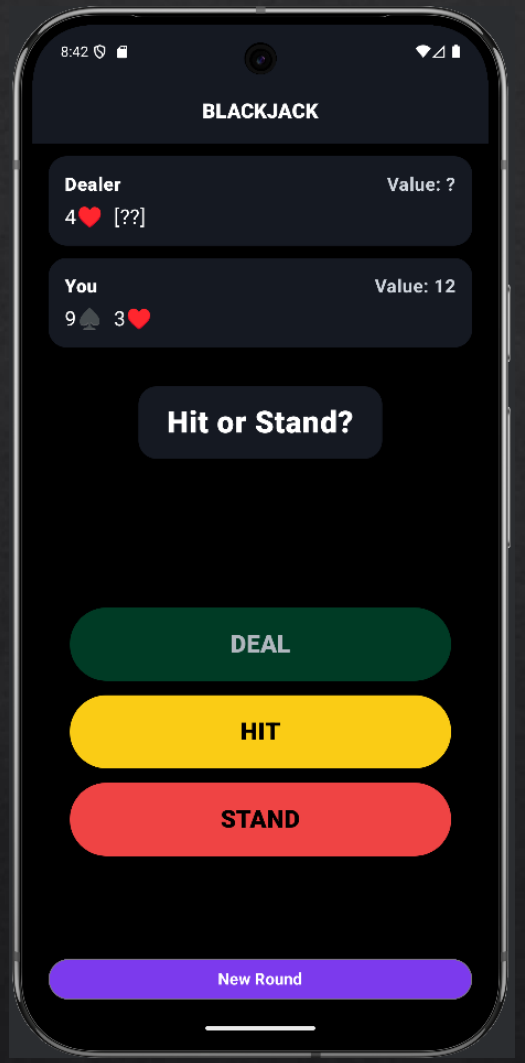


◈ Player Turn

Player chooses between **Hit** and **Stand**

Dealer’s second card is hidden

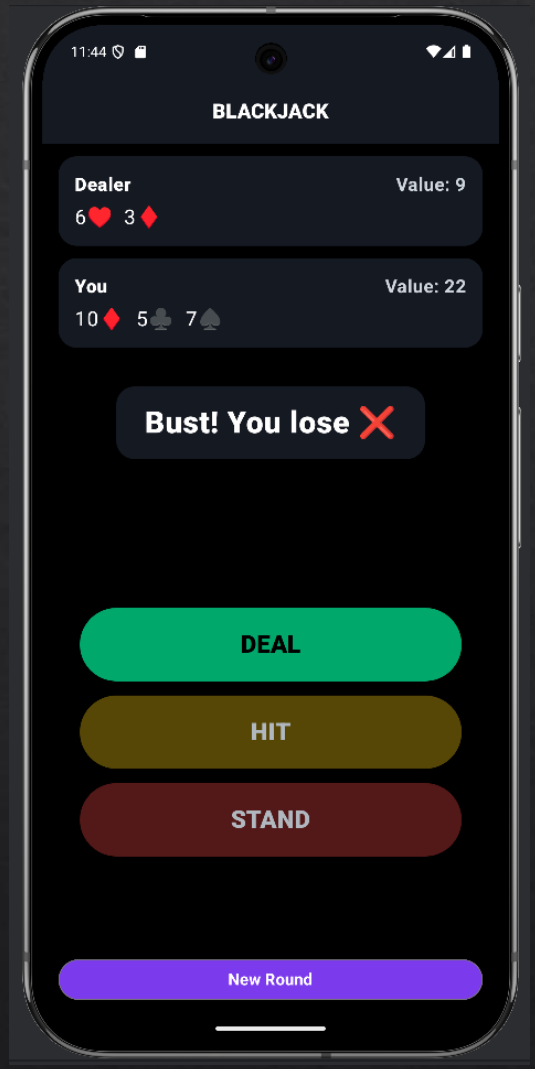
Current hand values are displayed

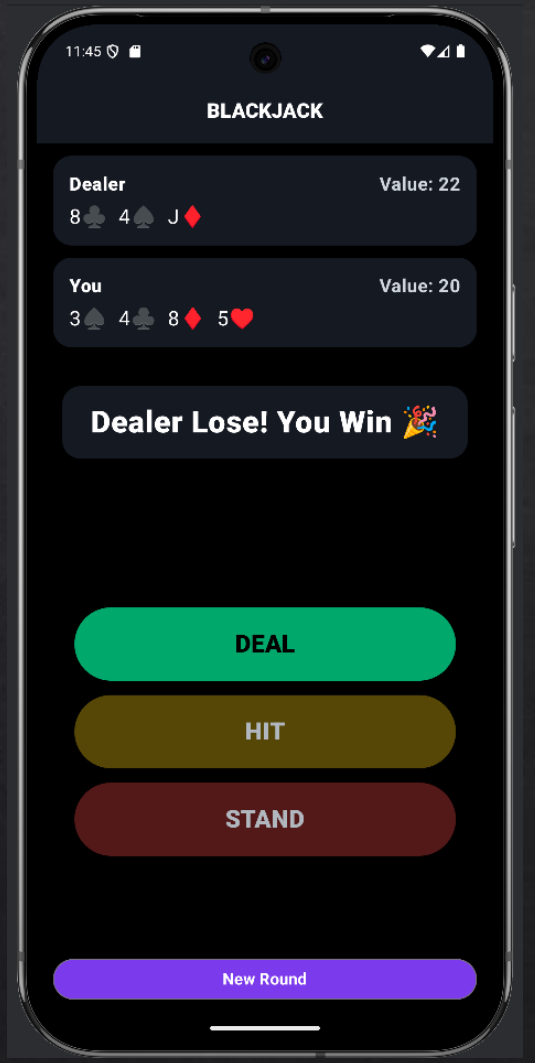


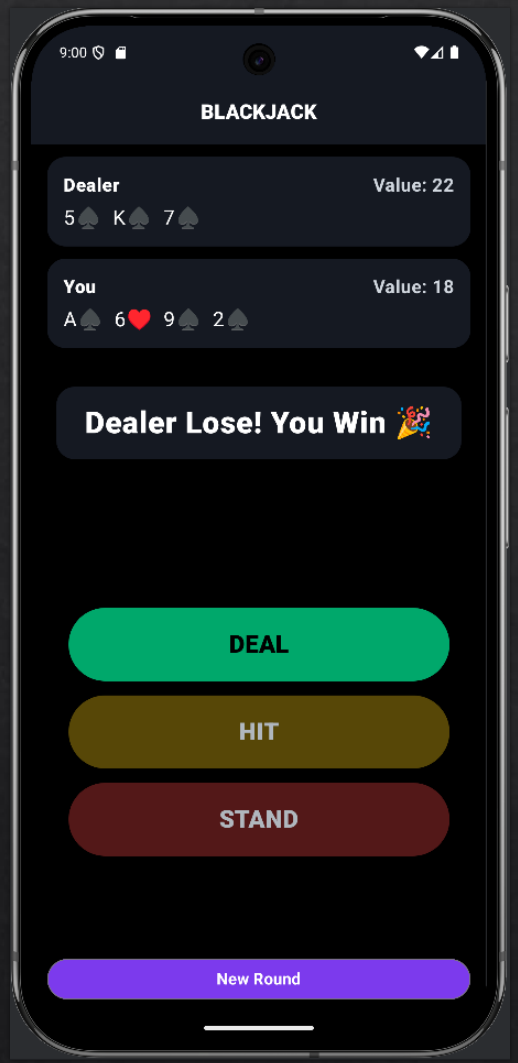
◈ Game Result

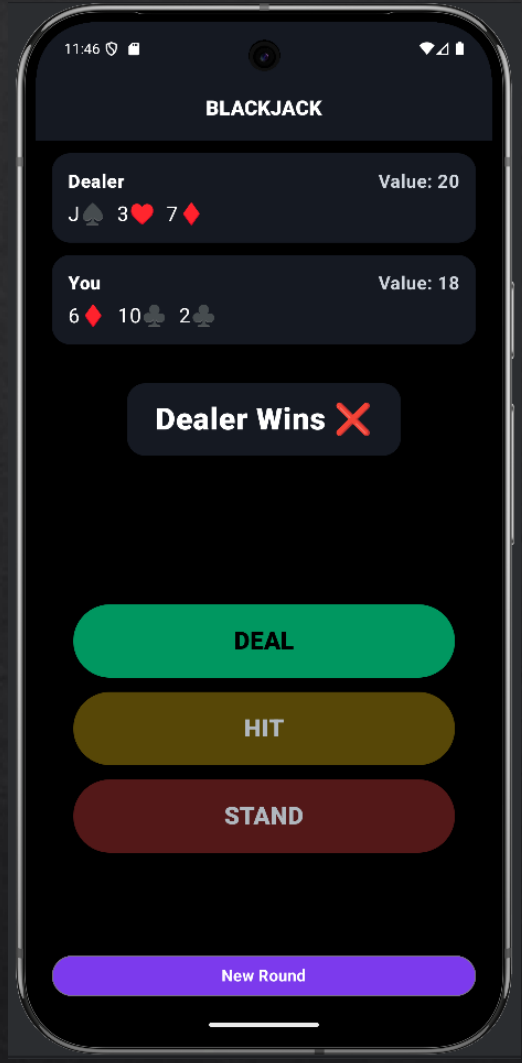
Final result screen after the round ends

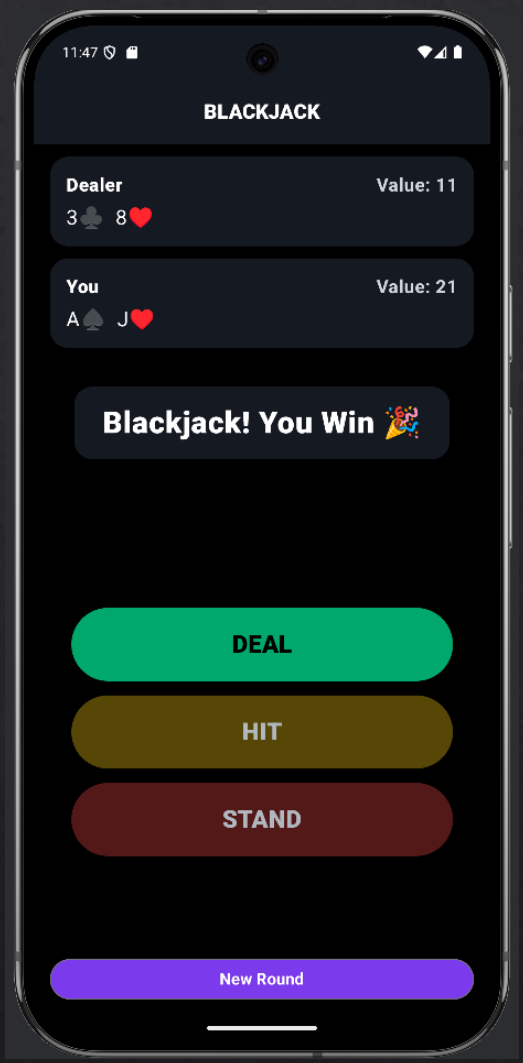
Possible outcomes: **Win, Lose, Push, Blackjack**

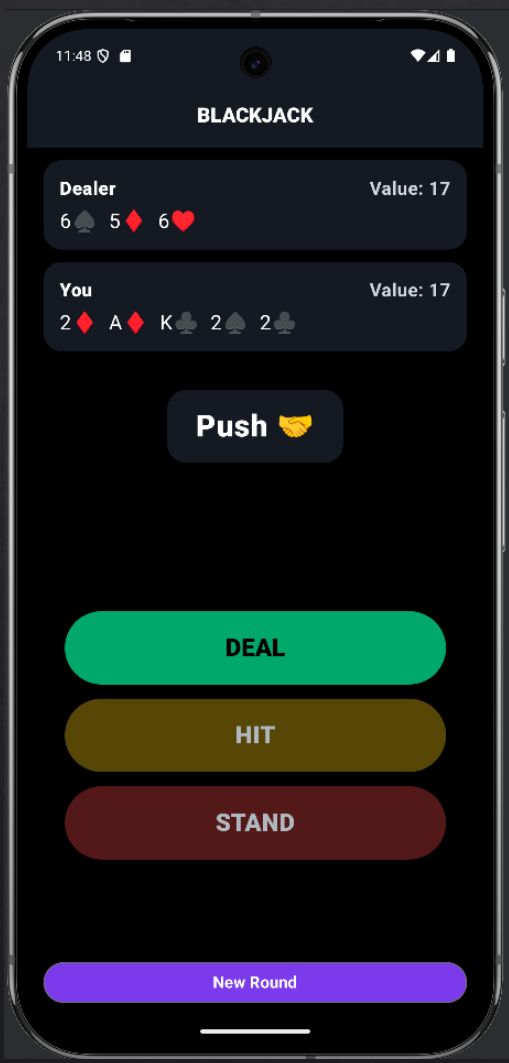


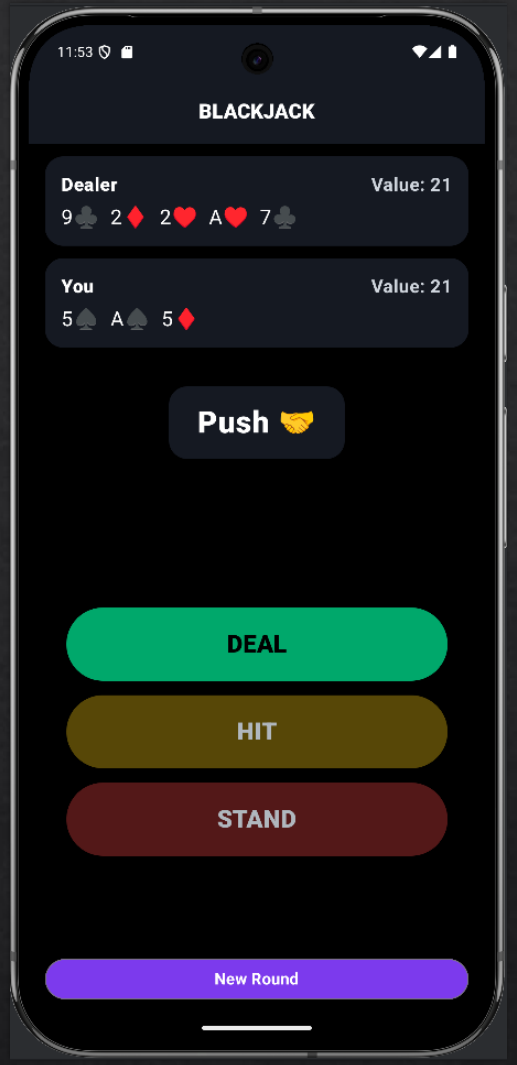


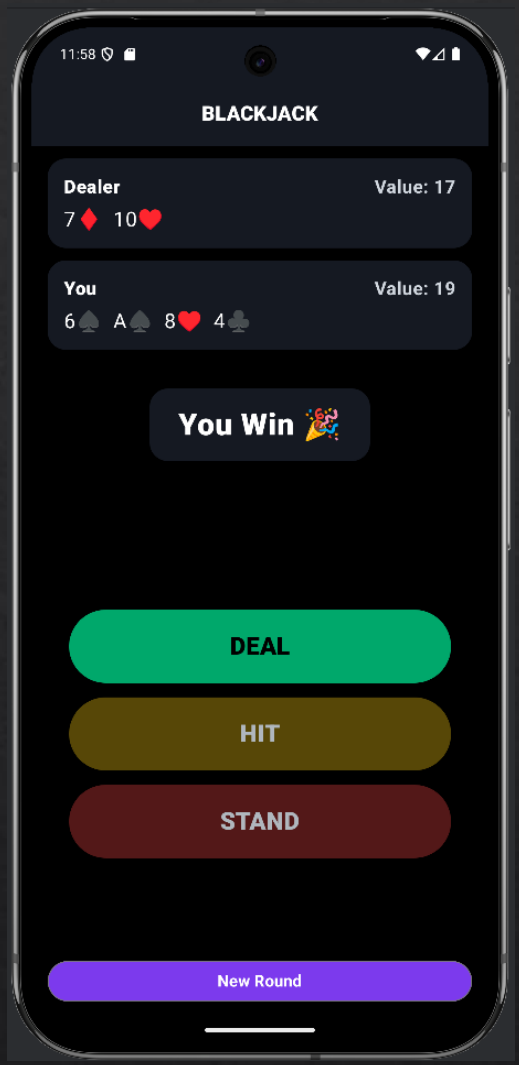

9:00
BLACKJACK
Dealer
Value: 22
5♠ K♠ 7♠
You
Value: 18
A♠ 6♥ 9♠ 2♠
Dealer Lose! You Win 🎉
DEAL
HIT
STAND
New Round

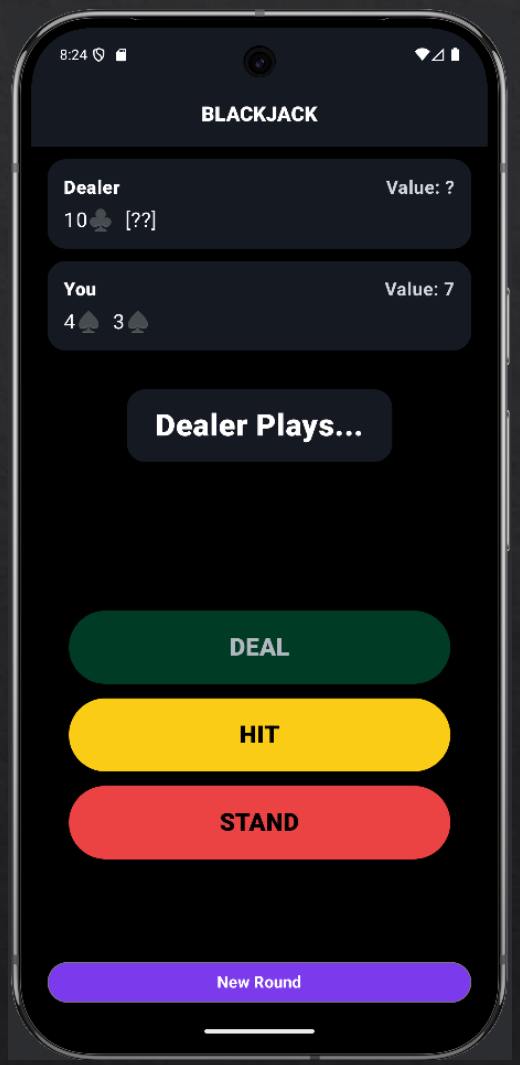

11:46
BLACKJACK
Dealer
Value: 20
J♠ 3♥ 7♦
You
Value: 18
6♦ 10♣ 2♣
Dealer Wins ❌
DEAL
HIT
STAND
New Round


11:47
BLACKJACK
Dealer
Value: 11
3♣ 8♥
You
Value: 21
A♠ J♥
Blackjack! You Win 🎉
DEAL
HIT
STAND
New Round


11:48
BLACKJACK
Dealer
Value: 17
6♠ 5♦ 6♥
You
Value: 17
2♦ A♦ K♣ 2♠ 2♣
Push 🤝
DEAL
HIT
STAND
New Round


11:53
BLACKJACK
Dealer
Value: 21
9♣ 2♦ 2♥ A♥ 7♣
You
Value: 21
5♠ A♠ 5♦
Push 🤝
DEAL
HIT
STAND
New Round


11:58
BLACKJACK
Dealer
Value: 17
7♦ 10♥
You
Value: 19
6♠ A♠ 8♥ 4♣
You Win 🎉
DEAL
HIT
STAND
New Round


8:24
BLACKJACK
Dealer
Value: ?
10♣ [??]
You
Value: 7
4♠ 3♠
Dealer Plays...
DEAL
HIT
STAND
New Round

◈ Conclusion

A functional Blackjack mobile game was successfully developed for the Android platform.

The application follows real Blackjack rules and provides an interactive user experience.

Jetpack Compose enabled a modern, clean, and efficient user interface.

The project improved understanding of mobile application development and game logic.

◈ Future Improvements

Implementing **animations and sound effects**

Supporting **multiple languages**

Enhancing dealer AI and game difficulty

Optimizing the application for tablet and landscape mode (currently optimized only for a 6.8-inch portrait-oriented device).

Thank You for Your Attention

Any Questions?